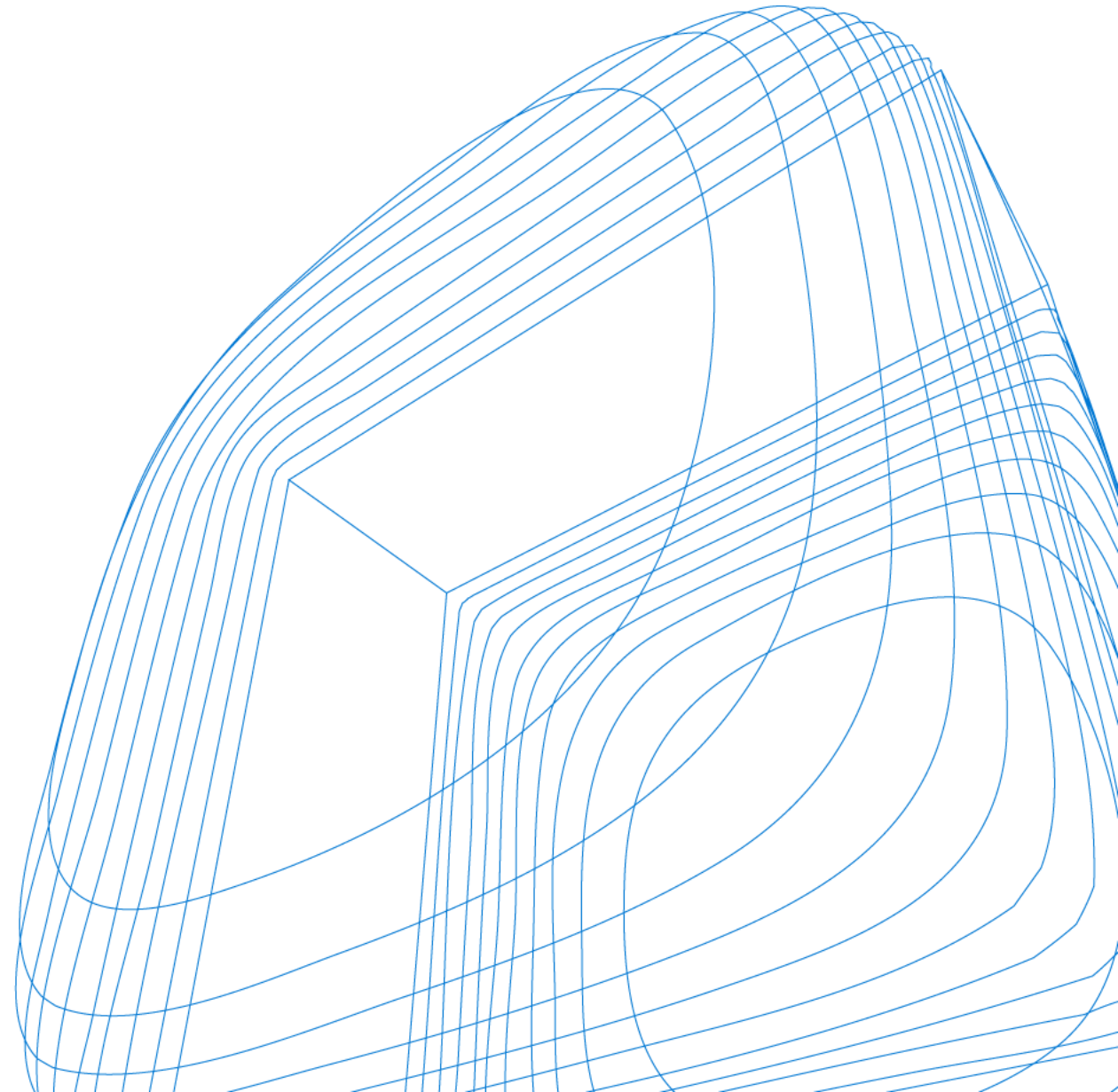
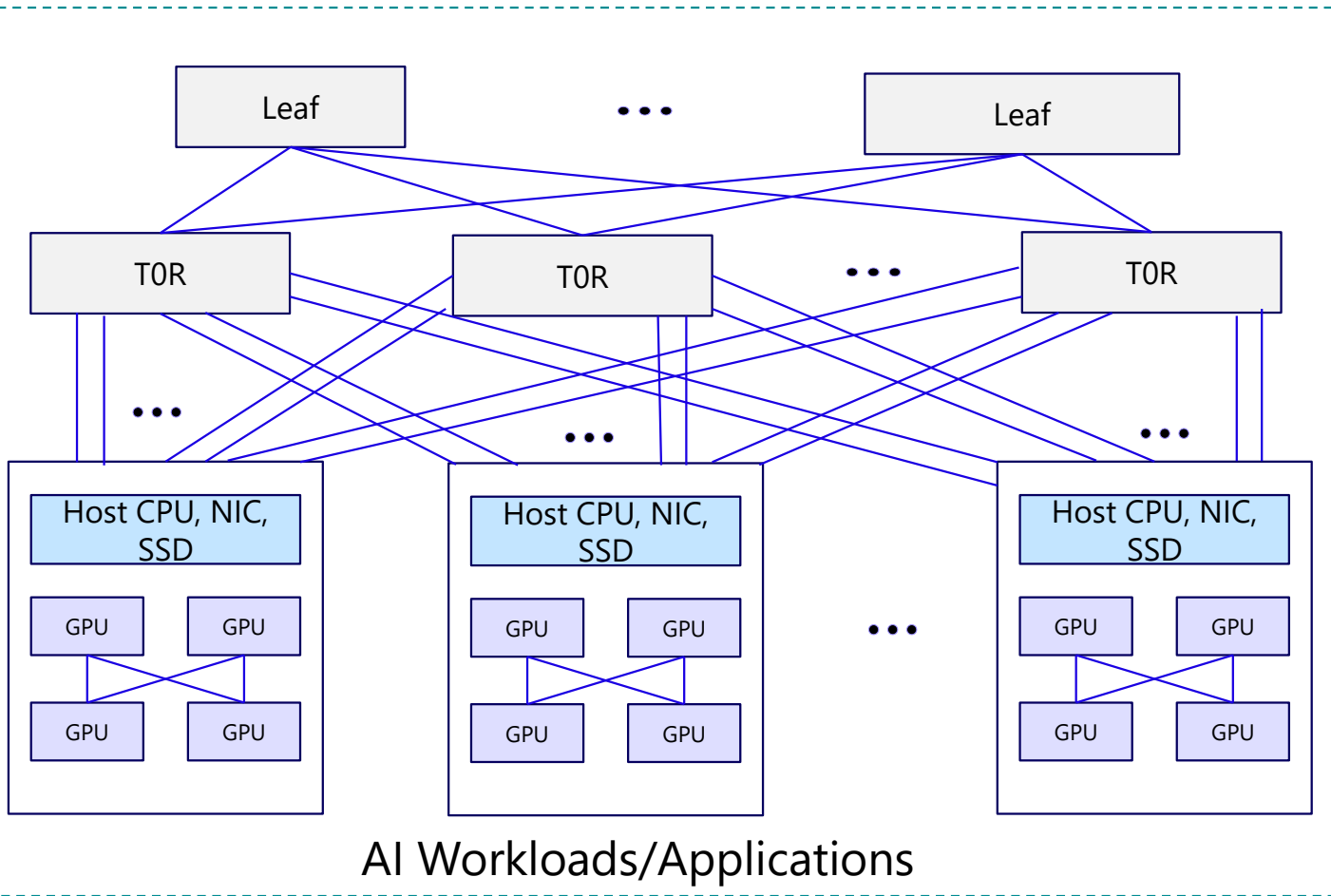


SRv6 in Microsoft Fairwater AI Backend Network

Changrong Wu, Microsoft



Artificial Intelligence in the Cloud



New Traffic Pattern:

- Small number of large flows
- Periodic bursts of data sent synchronously

New Requirements:

- Demand for finer-grained network control

Continental-scale GPU Cluster Emerging



From Wisconsin to Atlanta: Microsoft connects datacenters to build its first AI superfactory

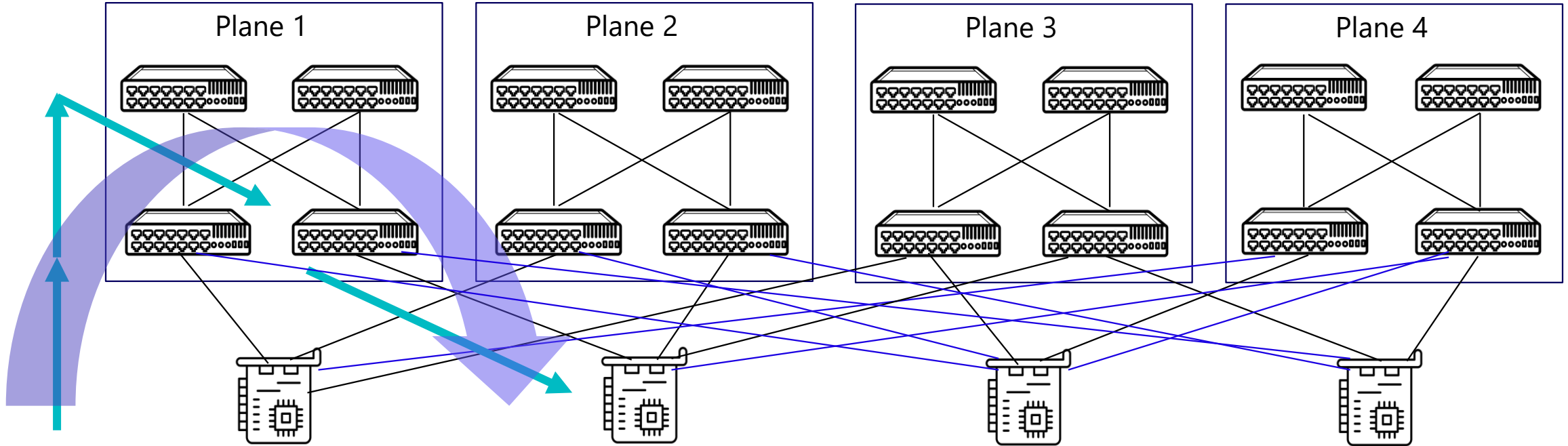
Geo-distributed Hyperscale Datacenter Network is rising

- Power Supply, Physical Space, etc. limit the scale of a single DC site.
- The demand of GPU capacity from a single job is set to grow beyond the capacity of a single DC.

New Challenges for Networking

- All features must be cost-effective and scalable
 - No Proprietary Technology
- Traditional passive hash-based load balancing mechanisms suffered from low entropy problem.
 - Need more active traffic engineering
- Failures are inevitable at this scale
 - Fast failover is necessary
- Multi-path transport is desired for efficient bandwidth utilization
 - Explicit support of multi-path packet spraying

Fairwater AI Backend Network with SRv6



- Provides fined-grained network control based on source routing
- Enables path enumeration for traffic management
- Integration with AI workloads flow scheduling provides optimal network performance
- Allow source to quickly reroute upon path failures or congestion

Support a Wide Range of Endpoint Transport Options

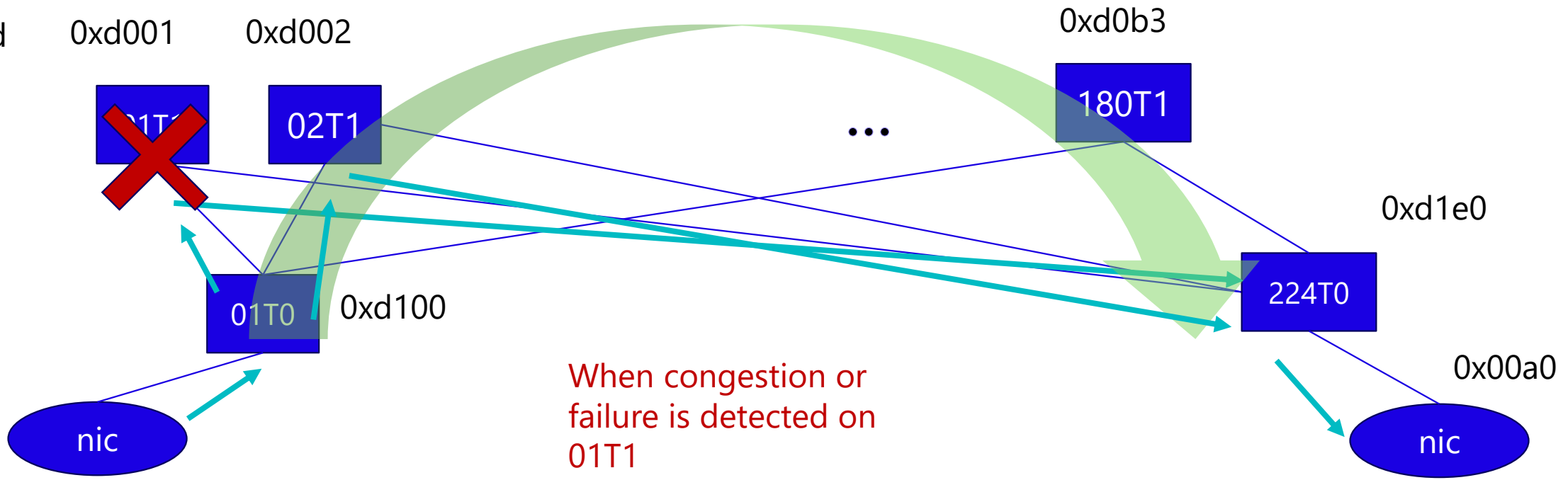
Multi-path transport protocols that benefit from SRv6

- Multipath Reliable Connection (MRC)
- UCCL-Tran
- Falcon
- MetaROCE
- ...

Compatible with major congestion signals

- Packet trimming
- ECN marking
- Inband Network Telemetry

SRv6 with uN SID



DstIPv6: *fcbb:bbbb:d100:d001:d1e0:00a0::*

NO SRH

1st hop 2nd hop 3rd hop 4th Hop

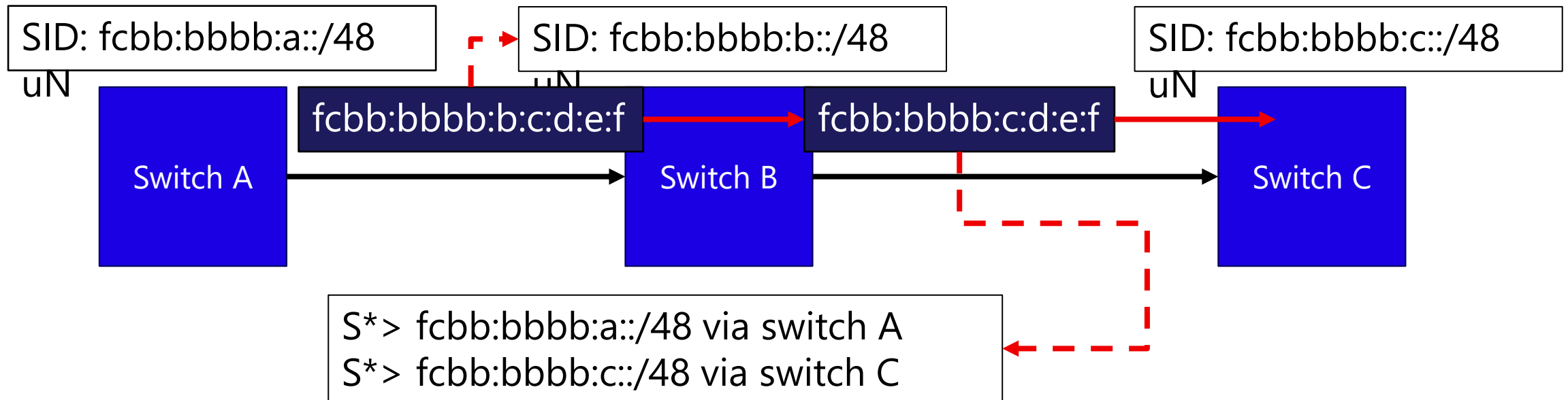


fcbb:bbbb:d100:d002:d1e0:00a0::

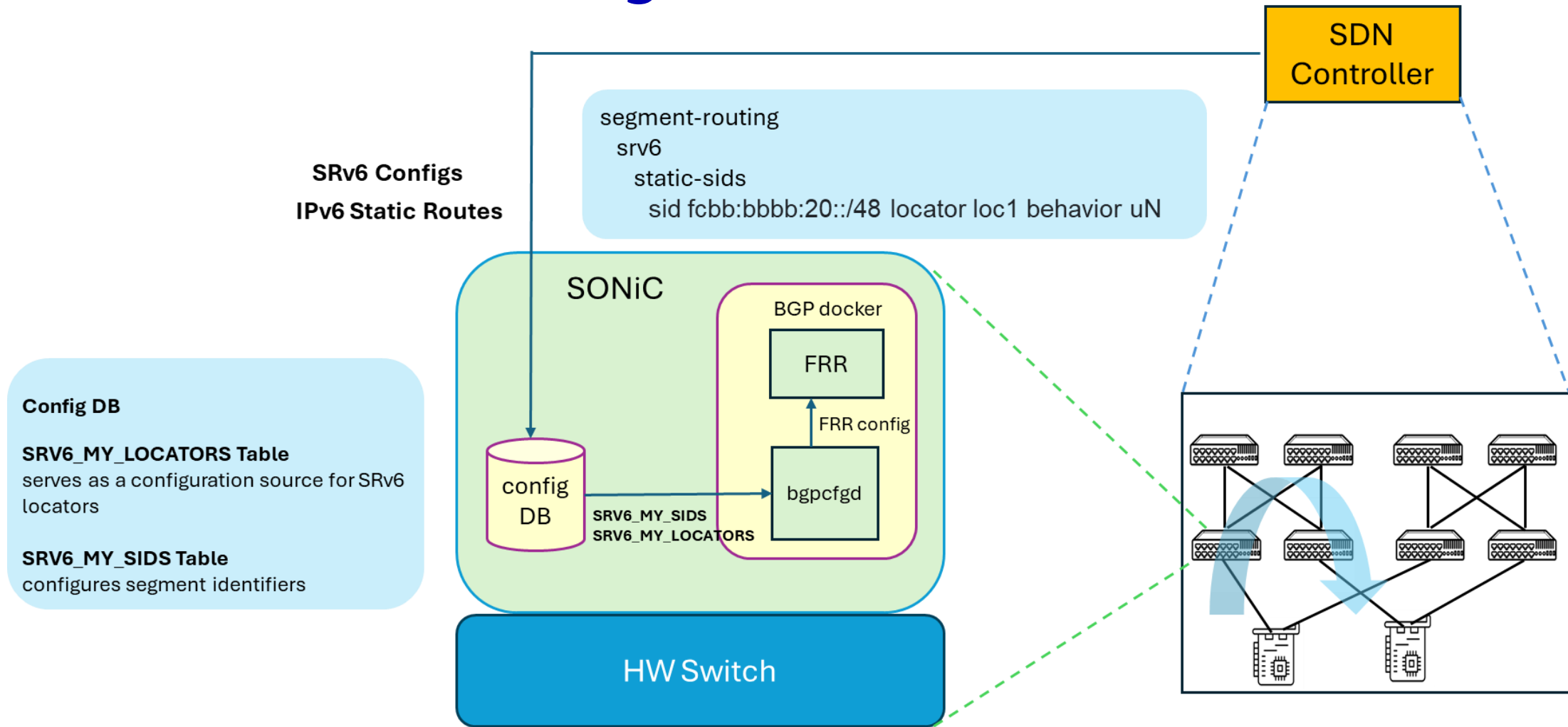
NIC switches to new path by change the uSIDs encoded in the packet header

Simplified static SRv6 control plane

- Segment Identifiers(SIDs) are configured on switches statically.
- The switch has a static route configured for each of its neighbor's SIDs.
- Hosts get the list of SRv6 paths (encoded as a list of SIDs) that it should use from the central controller.



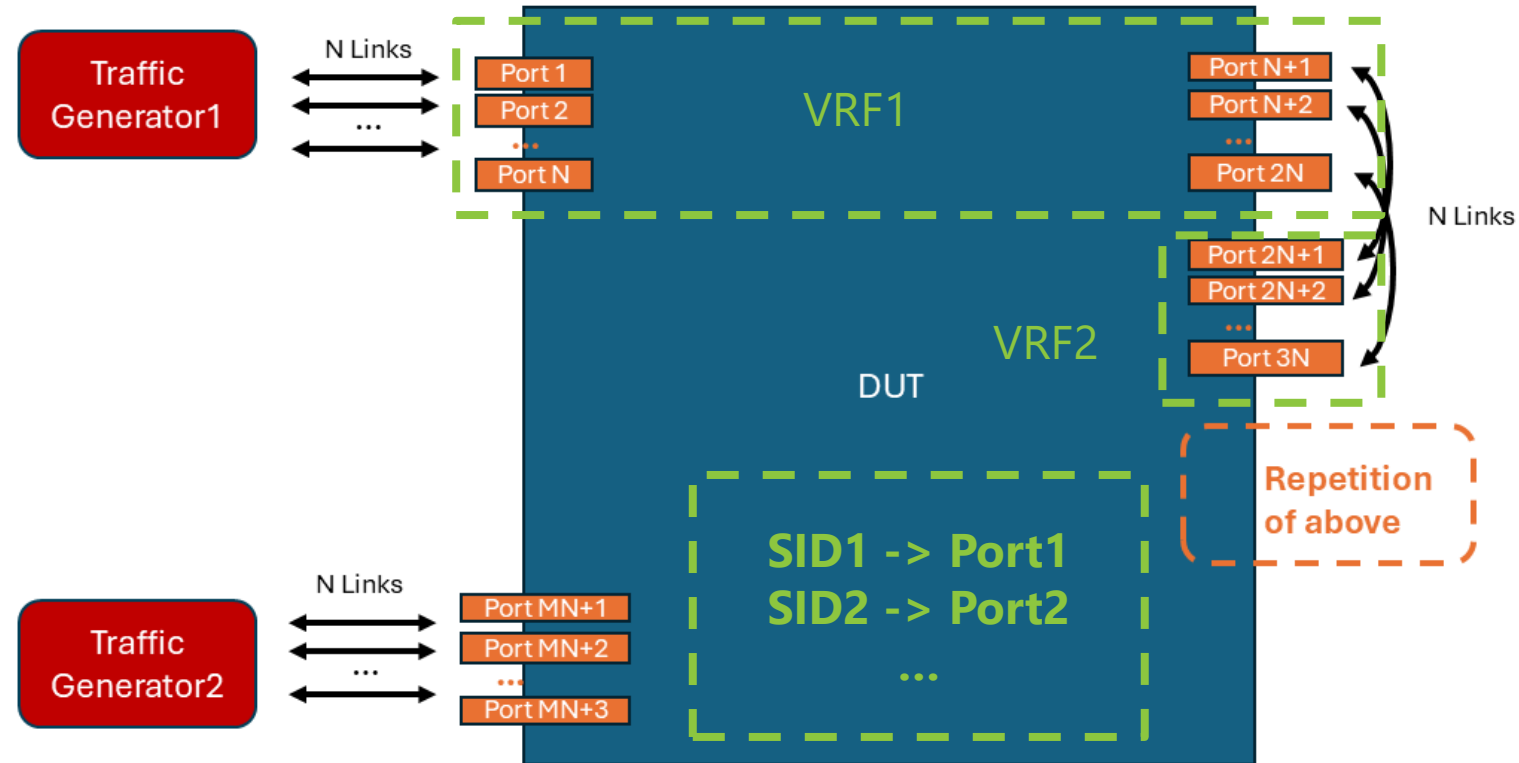
SRv6 static uSID config in SONiC



Stress Testing Switch ASIC with SRv6

Use snake topology to maximize traffic stress on the ASIC

- Use VRFs to avoid routing domain collision
- Use multi-SIDs to allow packets selecting different interfaces as egress ports
- (optional) Use VLAN sub-interfaces to support bidirectional traffic



Warmboot Support for SRv6 Static SIDs in SONiC

LLM training jobs are long-running workload on valuable GPU resources, which are not supposed to be interrupted.

- Image upgrade based on hitless warmboot is necessary for fast maintenance.
- Coming in SONiC 202611 release
 - Already supported on some platforms
 - More to be announced
- Achieved parity with conventional route objects
 - Reconciliation in Fpmsyncd and Syncd
 - Simpler to implement since uN SIDs are stateless

SRv6 Link/Switch Isolation

Traditional livesite mitigation relies on using BGP Shut/unshut as well as TSA/TSB route-map to isolate faulty components.

- How to isolate link or switch without BGP?

In Fairwater AI Backend Network, we use extremely efficient ACLs

- Defined a lite ACL table type that consumes minimal length of TCAM field
- Preloaded the dedicated ACL table/rules on the device without binding to any port
- Use optimized SONiC GCU to change port bindings per isolation/unisolation needs

SRv6-based Network Scanning

Traditional Network Scanning approach: IPinIP-based multi-layers probing

- Netbouncer (NSDI'19)
- Probabilistic inference since no deterministic path control
- Large volume of probes needed for high-accuracy result

New approach with SRv6: uSID steering-based single layer probing

- 6 uSIDs can steer probe on a round trip across 2-tiers network
- Deterministic path control enables deterministic localization

Thank You!

